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Step 6 - Physical Counter (CNTPCT)

Reference Note

This chapter is based on the ARM Generic Timer Architecture.

The Physical Counter provides software with a direct view of the System Counter.

Learning Objectives

After this chapter you should understand:

- Purpose of CNTPCT
- Relationship to System Counter
- Reading physical time
- Counter behavior

1. What is CNTPCT?

`CNTPCT` provides the Physical Counter value.

Conceptually:

Source	View
System Counter	CNTPCT

2. What Does It Return?

Example:

MRS X0, CNTPCT

Result:

1,000,000

Meaning:

The System Counter currently holds
1,000,000 ticks.

3. Live Counter Behavior

The counter never stops.

Example:

MRS X0, CNTPCT
MRS X1, CNTPCT

Possible result:

Register	Value
X0	1,000,000
X1	1,000,012

Reason:

Time advanced between reads

4. Measuring Elapsed Time

Procedure

Step	Operation
1	Read counter
2	Execute operation
3	Read counter again
4	Compute difference

Formulas:

Elapsed Ticks
=
End - Start
Elapsed Time
=
Elapsed Ticks / CNTFRQ

5. Why Physical Counter Matters

The Physical Counter is the foundation for:

- Time measurement
- Timer programming
- Timeout calculations
- Virtualization

Key Takeaways

Concept	Description
CNTPCT	Physical counter register
Source	System Counter
Behavior	Always increasing
Purpose	Measure time

Knowledge Check

Q1

What does CNTPCT represent?

Q2

Why can two consecutive reads differ?

Q3

How is elapsed time calculated?

Answer Key

Question	Answer
Q1	Current physical counter value
Q2	Counter continued advancing
Q3	End Count – Start Count